PROJECT PROPOSAL
ON

Hospital Management System

Guided By:-

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Batch code- ANP-D2405

2. Introduction

The Hospital Management System is designed to manage patient records, doctor details, staff
information, appointments, and billing. It helps in maintaining structured healthcare data and
improves the efficiency of hospital operations.

3. Objective

- To digitize patient data management.
- To schedule appointments efficiently.
- To manage billing records.
- To store and retrieve doctor and staff details securely.

4. Project Category

Database Management System (DBMS) / Java Application

5. Analysis

Modules and Description

Patient Module:
Stores patient details such as Name, DOB, Gender, Phone Number, Patient ID,
and Disease.

Doctor Module:
Manages doctor details including Name, Specialization, Phone Number, and Doctor ID.

Staff Module:
Contains Staff details like Name, Role, Phone Number, and Staff ID.

AppointmentModule:
Handles scheduled appointments using Patient ID, Doctor ID, Date, and Time.

Billing Module:
Stores billing details including Patient ID, Appointment ID, Amount, and BillingDate

Admin Table:
The Admin table stores login credentials and role details of system administrators to manage hospital operations.
It ensures secure access control and allows authorized users to maintain system data efficiently.

Lab Report Table:
The Lab Report table maintains patient test details, results, and diagnostic observations recorded by laboratory staff.
It helps track medical tests and provides accurate reports for patient diagnosis and treatment.

Modules and Description

Blood Bank Management System :

The Blood Bank Management module manages blood donor records, blood group inventory, and availability status in real time. It ensures smooth blood request processing, donation tracking, and timely allocation during medical needs.

Emergency Service Management :

The Emergency Service Management module handles emergency patient registration, ambulance coordination, and rapid response operations. It ensures immediate medical attention and efficient communication between emergency staff and hospital units.

Database Design - Tables

Admin table:

Column	Datatype	Description
admin_id	INT	Admin's unique identifier
admin_name	VARCHAR(100)	Admin's name
admin_email	VARCHAR(255)	Admin's email address
admin_password	VARCHAR(255)	Admin's password

Doctor Table:

Fields	Data Type	Description
doctor_id	INT	PRIMARY KEY
name	VARCHAR(100)	Doctor name
specialization	VARCHAR(100)	specialty field
phone	VARCHAR(100)	contact number

Patient Table:

Fields	Data Type	Description
patient_id	INT	PRIMARY KEY
name	VARCHAR(100)	patient full name
dob	DATE	Date of birth
gender	VARCHAR(100)	Gender
phone	VARCAHR(100)	Contact Number
disease	VARCHAR(100)	Disease details

Database Design - Tables

Staff Table:

Fields	Data Type	Description
staff_id	INT	PRIMARY KEY
name	VARCHAR(100)	Staff name
role	VARCHAR(100)	Designation
phone	VARCHAR(15)	Contact Number

Lab Report Table:

Fields	Data Type	Description
report_id	INT	Lab report unique identifier
patient_id	VARCHAR(100)	Patient associated with the report
test_name	VARCHAR(100)	Name of the test conducted
test_result	VARCHAR(255)	Result of the test

Database Design - Tables

Blood Bank table

Fields	Data Type	Description
blood_id	INT	PRIMARY KEY
blood_group	VARCHAR(5)	Blood group(A+, O_,etc)
quantity	INT	Available units
doctor_id	INT	Linked donor
collection_date	DATE	Date of blood

Emergency Service table

Fields	Data Type	Description
incident_id	INT	PRIMARY KEY
incident_type	VARCHAR(5)	Type of incident
location	VARCHAR(10)	Incident location
response_time	DATETIME	Time of response
status	VARCAHR	status of incident

Database Design - Tables

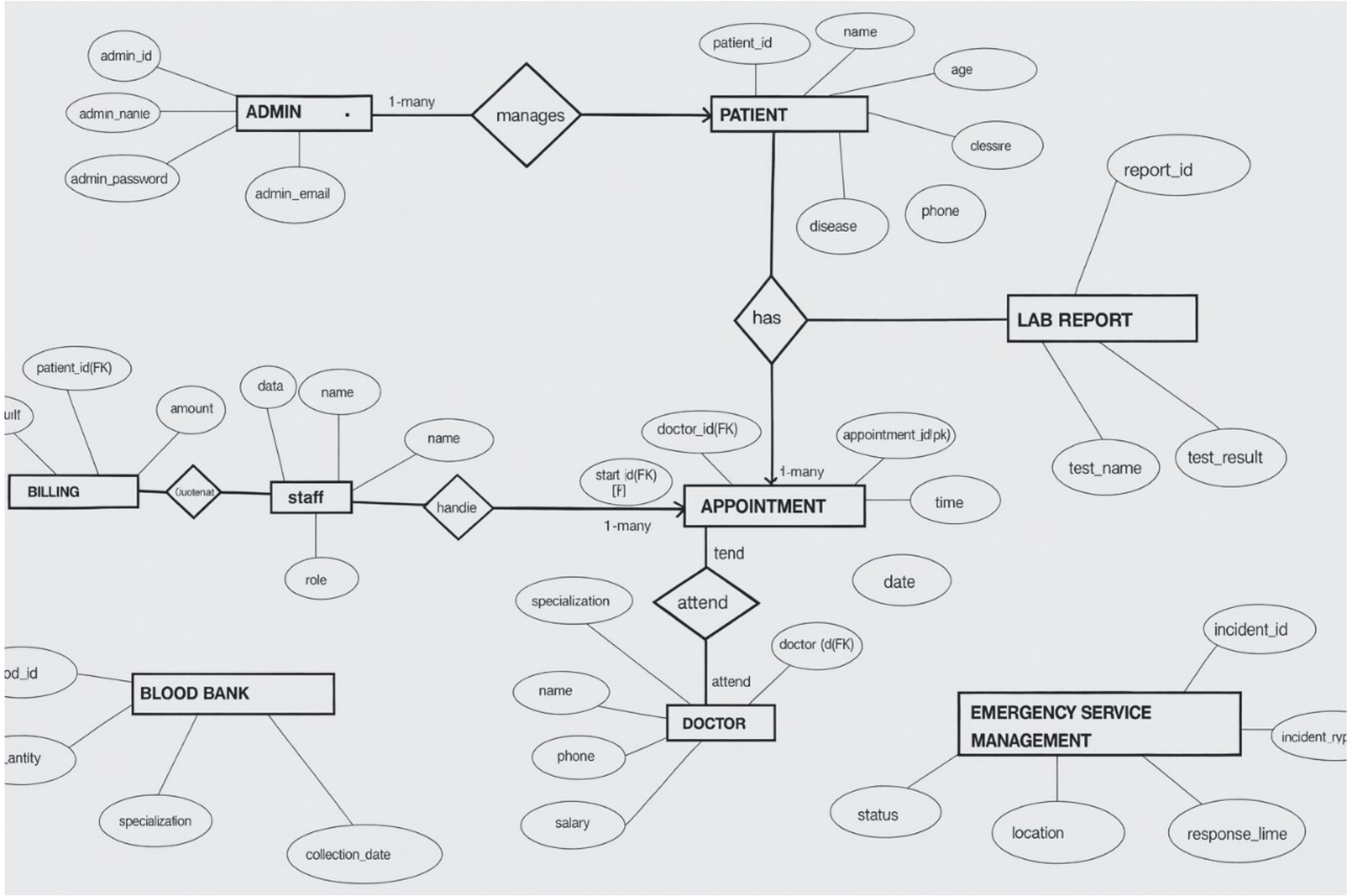
Appointment table

Fields	Data Type	Description
appointment_id	INT	PRIMARY KEY
patient_id	INT	FOREIGN KEY
doctor_id	INT	FOREIGN KEY
Date	DATE	appointment date
time	TIME	appointmnet time

Billing table

Fields	Data Type	Description
bill_id	INT	PRIMARY KEY
patient_id	INT	FOREIGN KEY
appointment_id	INT	FOREIGN KEY
amount	DECIMAL(10,2)	total bill amount
date	DATE	billing date

ER Diagram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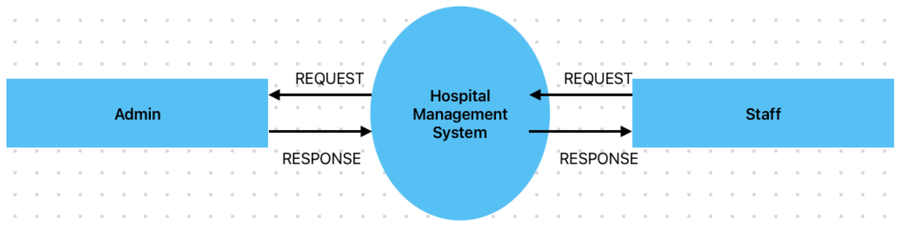
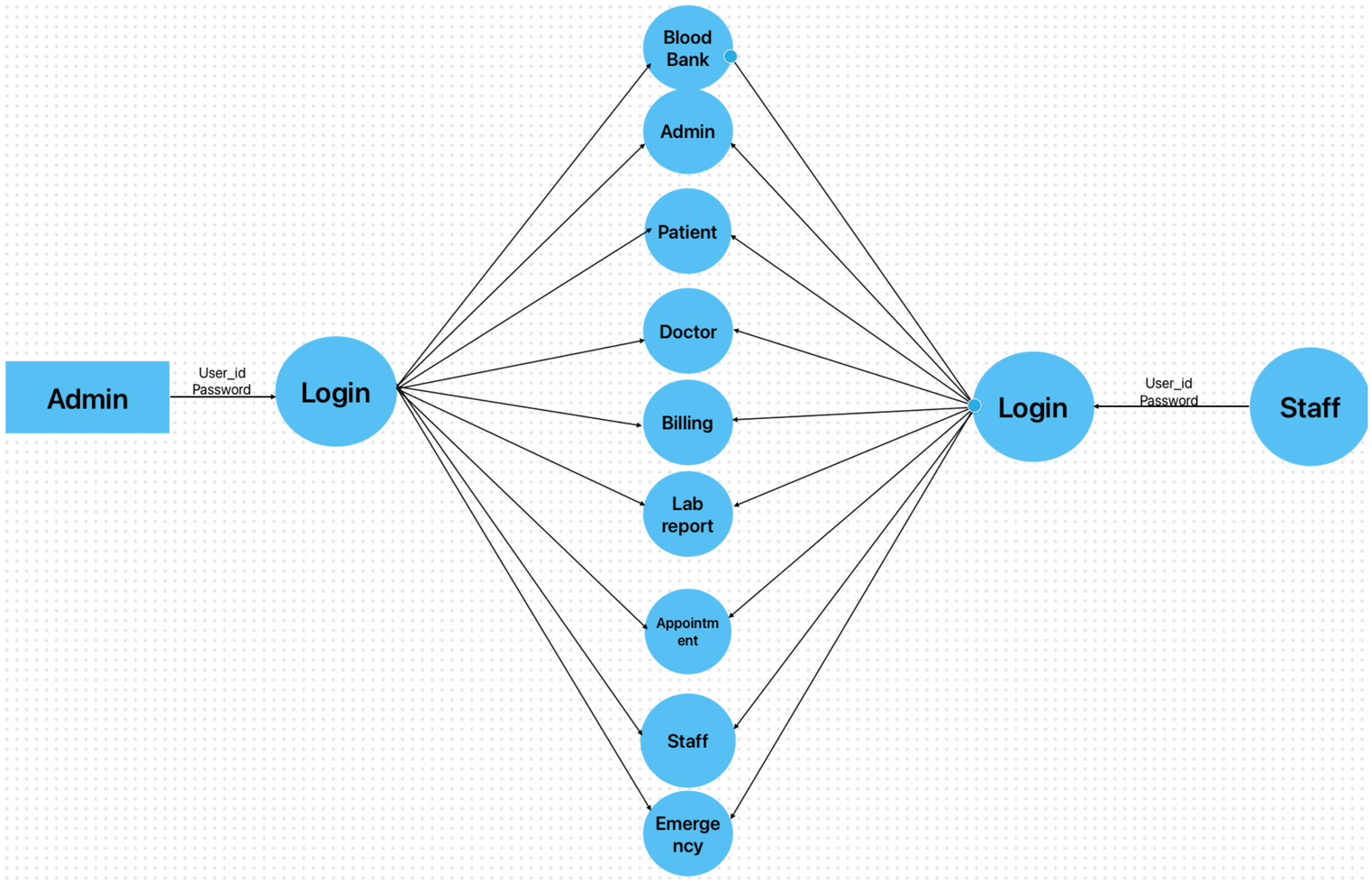
Data Flow Diagram

DFD Level 0

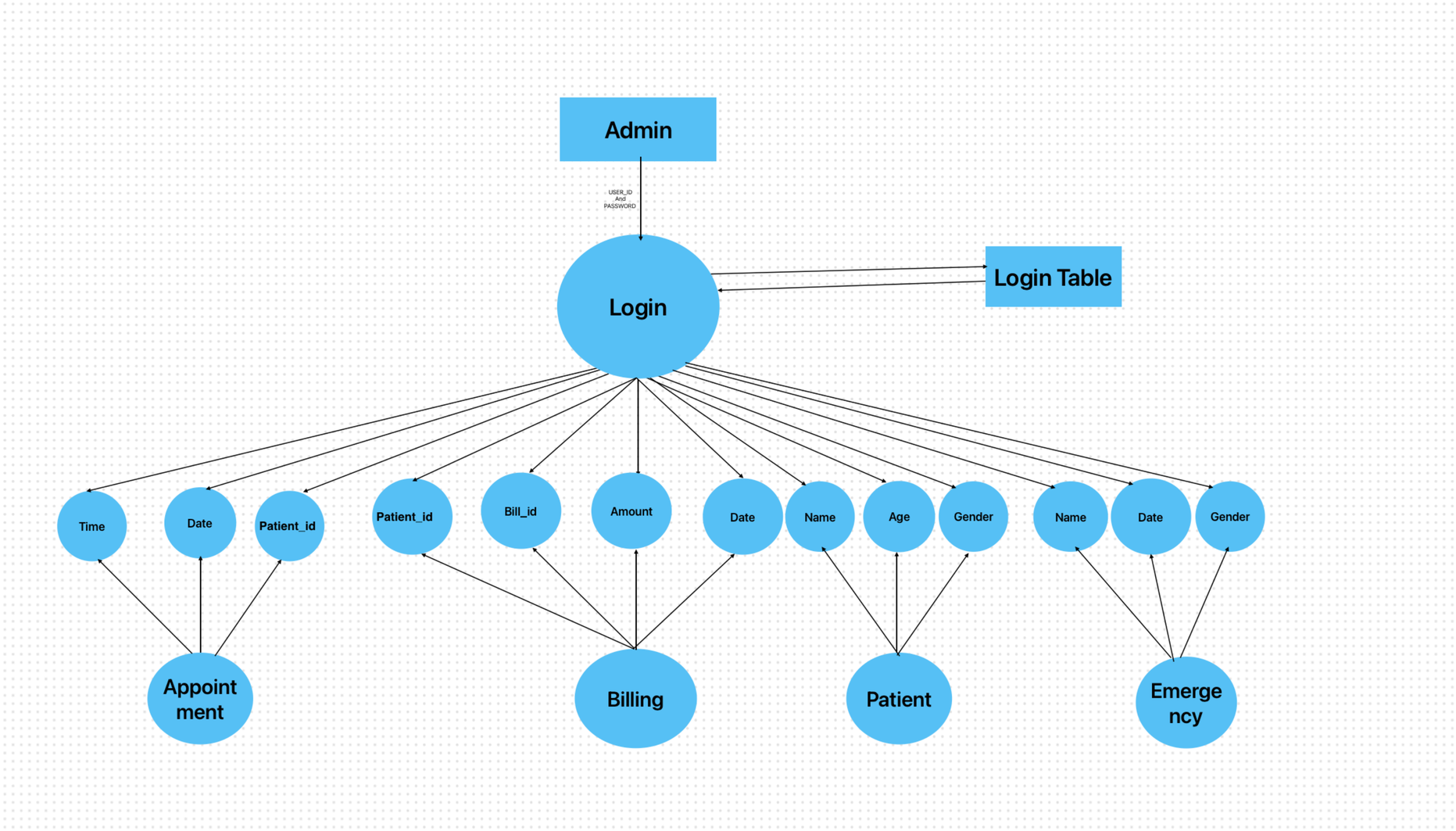


DFD Level 1



Data Flow Diagram

DFD Level 2



7. Platform Used

Hardware Requirements:

- PC with minimum 4GB RAM
- 500MB storage

SoftwaRequirement:

- Java JDK
- Eclipse IDE
- MySQL Server
- JDBC Driver

8. Future Scope

. Can be upgraded to include pharmacy management.

• Can integrate cloud-based storage.

• Can implement patient monitoring dashboard.

9. Bibliography

• Java Complete Reference

• MySQL Documentation

• JDBC Official Guide